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20ECD1007
Section-4 Grp-1

Malus' Law

Date of experiment: **Partner's name:**

Date of Lab. report submission:

Aim of the Experiment

To verify Malus' Law of polarization of light.

Apparatus Required

(i) Optical bench.
(ii) Laser light source.
(iii) Polarizer with stem for mounting.
(iv) Photometer (Optical base plate).
(v) Digital multimeter with connecting wire.

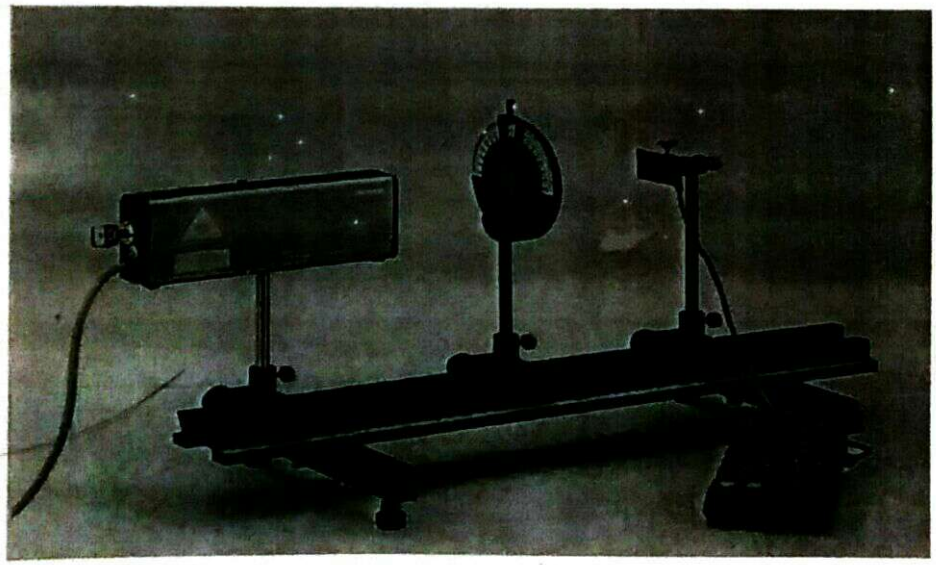


Fig-1: Experimental setup

Theory

Figure-1 depicts the complete experimental set up for verification of Malus' law of polarization. When linearly polarized light passes through a Polaroid, the intensity of the transmitted light varies with the angular position of the Polaroid w.r.t. the angle of the plane of polarization of incident light. The emergent light

Procedure

(i) Place the set-up is in a dark room so that excess light except the laser source may be avoided.

(ii) Adjusted the source of light, Polaroid and the photometer on the optical bench so that the light after passing through the Polaroid falls at the centre of the photometer.

(iii) Before the laser is switched on note down the effect of the stray light on the photometer which will be finally deducted from the observed values.

(iv) Switch on the laser source for 30 minutes and record the signal from the digital multi-meter for each specific positions of the Polaroid.

(v) Plot graph between the square of the cosine of the polarising angle vs. the photocurrent (I) from 0^0 to 90^0 and back to 0^0.

(vi) Plot the graph between the differential current $I'-I_0$ and θ.

Observations

Table-1: Table for current

Sl. No.	Stray current $I_S(\mu A)$	θ	$Cos^2\theta$	Photocurrent $I(\mu A)$	$I-I_S(\mu A)$ $=I'$	Difference Current $I'-I_0(\mu A)$
1	0.0	90°	0	14.8	14.8	-171.9
2	0.0	80°	0.03	12.0	12.0	-174.7
3	0.0	70°	0.117	10.0	10.0	-176.7
4	0.0	60°	0.25	16.7	16.7	-170
5	0.0	50°	0.413	27.7	27.7	-159
6	0.0	40°	0.587	49.0	49.0	-139.7

Sl. No.	Stray current $I_S(\mu A)$	θ	$Cos^2\theta$	Photocurrent $I(\mu A)$	$I-I_S(\mu A)$ $=I'$	Difference Current $I'-I_0(\mu A)$
7	0.0	30	0.75	63.5	63.5	-123.2
8	0.0	20	0.883	76.0	76.0	-110.7
9	0.0	10	0.97	102.7	102.7	-84
10	0.0	0	1	129.6	129.6	-57.1
11	0.0	-10	0.09	147.5	147.5	-39.2
12	0.0	-20	0.883	144.1	144.1	-42.6
13	0.0	-30	0.75	118.5	118.5	-68.2
14	0.0	-40	0.587	113.8	113.8	-72.9
15	0.0	-50	0.413	85.9	85.9	-100.3
16	0.0	-60	0.25	70.4	70.4	-110.3
17	0.0	-70	0.117	53.5	53.5	-133.2
18	0.0	-80	0.03	31.8	31.8	-154.9
19	0.0	-90	0	20.6	20.6	-186.2

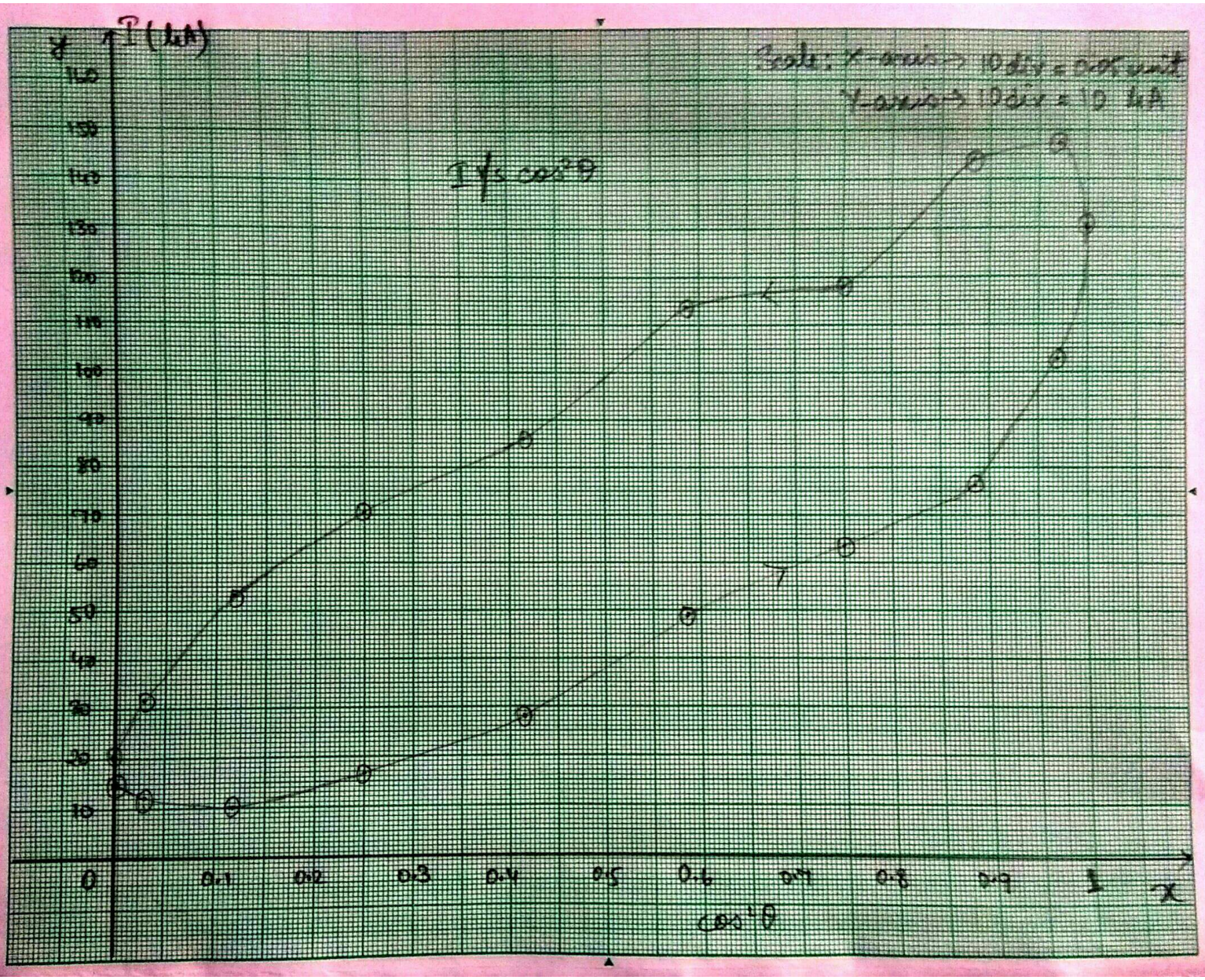
CONCLUSION :

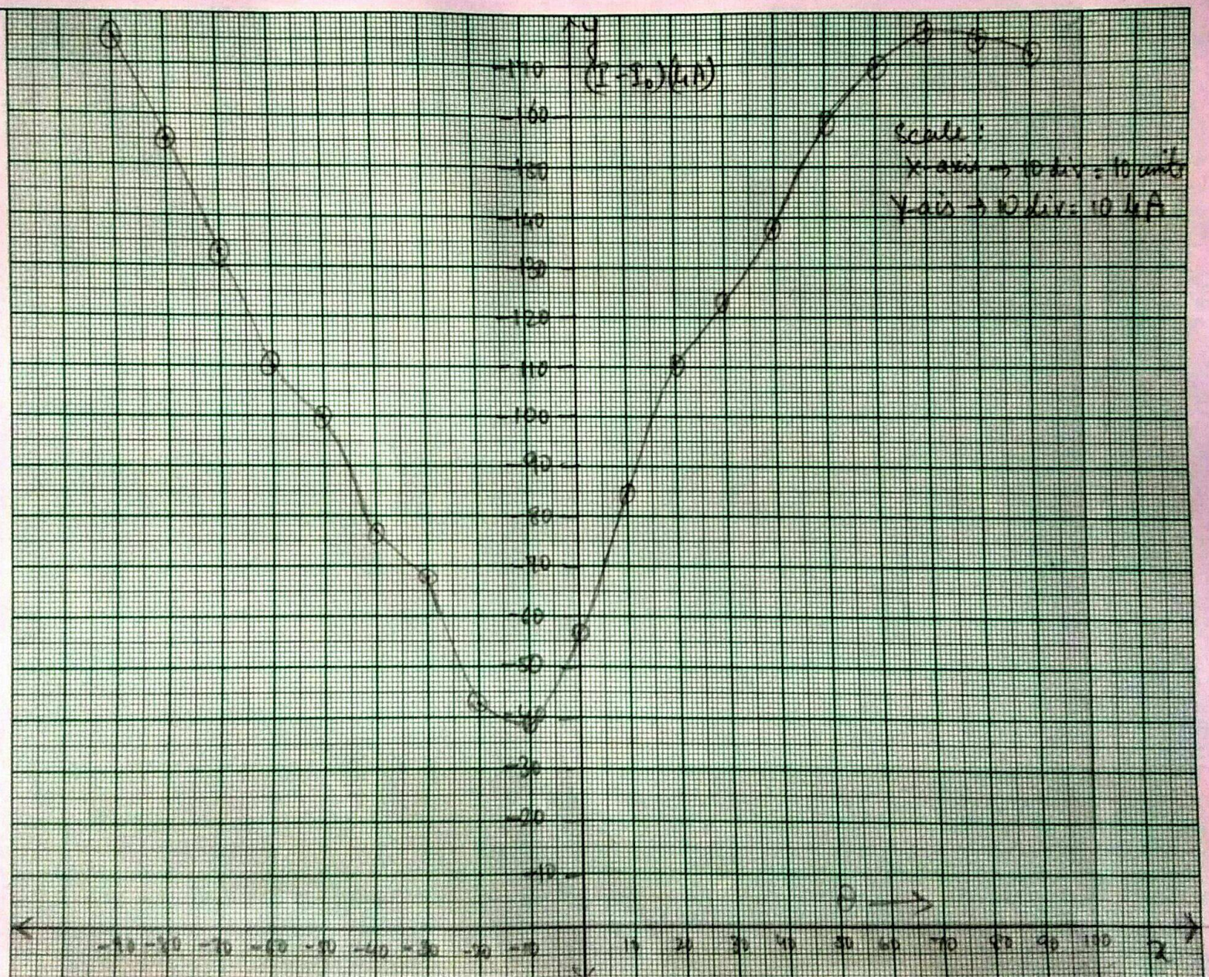
1.) Since the duration of values given by photocurrent is due to polarised light one can easily verify Malus' Law.

2.) The graph of $I' - I_0$ v/s θ is roughly similar to $\cos^2\alpha$, thus verifies Malus Law.

3.) The experiment should be conducted in a dark room to minimize the errors.

4.) Polariser's ~~ap~~ ability to change intensity is heavily used in making sunglasses and in photography.


I (kA)
Scale: X-axis → 10 div = 0.05 unit
Y-axis → 10 div = 10 kA
I vs cos²θ
y
x
cos²θ
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
10
20
30
40
50
60
70
80
90
100
110
120
130
140
150
160


y
(I - I₀)(μA)
170
160
150
140
130
120
110
100
90
80
70
60
50
40
30
20
10
θ
x
Scale:
X-axis → 10 div = 10 units
Y-axis → 10 div = 10 μA